



ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

Layers of cooperation: The interactional architecture of spoken language

Lecture 3

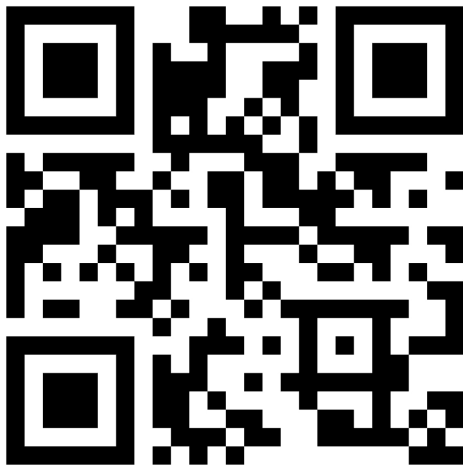
Emergent syntax and co-construction

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Speech Matters 2, Como, 19–22/5/2026

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Three lectures: the final step

Day	Lecture	Focus
19/5	Interaction as infrastructure	cooperative coordination, monitoring, reciprocity
20/5	Collaborative meaning-making	inferential cooperation, categorization, exemplification
22/5	Emergent syntax and co-construction	syntax across speakers, turns and interactional trajectories

»»➔ The final step is structural: how does the presence of the other enter the organization of syntax itself?

Lecture 3: co-construction and spoken UD

The lecture develops through five interconnected steps:

0. **From interaction to syntactic structure**
online syntax, projection, uptake and the theoretical status of co-construction
1. **Why spoken data are a problem for UD**
dependency trees, sentence boundaries, maximal units and speaker/dependency views
2. **Co-constructed syntax across speakers**
completions, parallel realizations, reformulations, extensions and cross-turn dependencies
3. **Related interactional phenomena**
backchannels, question-answer sequences, overlap, discourse markers and pauses
4. **Representing interactional architecture**
what must be preserved when co-constructed syntax is turned into annotated data

0/ From coordination to grammar

The previous lectures argued that spoken interaction is not merely a context in which language is used.

- ✓ participants continuously monitor one another
- ✓ meanings are stabilized through uptake, repair and alignment
- ✓ categories emerge through distributed inferential work

The next question is whether the same interactional pressure also shapes **syntactic organization**.

»»➔ If grammar is produced in time, the interlocutor is part of the conditions under which grammar emerges.

0/ Online syntax

Spoken syntax unfolds incrementally in real time.

Speakers continuously adapt syntactic trajectories to:

- ✓ addressee's feedback
- ✓ turn competition
- ✓ overlap and interruption
- ✓ repair trajectories
- ✓ epistemic and affective alignment

» Syntax should not therefore be seen as a finished product. It is a **process** co-constructed through, and shaped by, unfolding interaction.

(Auer, 2024; Helasvuoto, 2004; Lerner, 1991; Ono and Thompson, 1996; Sacks, Schegloff and Jefferson, 1974)

0/ Syntactic co-construction: projection and uptake

Syntax co-construction becomes possible because **utterances are not closed objects**. They project trajectories that others can recognize and act upon.



- ✓ syntax projects possible next actions
- ✓ recipients anticipate and display understanding
- ✓ the next contribution may complete, duplicate, reformulate or expand the structure

➡ The grammar we observe is the outcome of locally coordinated action.

(Auer, 2024; Lerner, 1991)



0/ What it looks like

A co-construction may be very small: one participant projects a syntactic path, and another participant completes it.

L his position is **pretty uh**
A **stable.**

BO146 ((ride)) vabbè te ti ci ti ci devo **fare pe::r**
BO139 ((ride)) **mestiere**

well I have to make you by
trade

»»» What looks like an incomplete fragment in one turn may become a complete syntactic relation across participants.

(Ono and Thompson, 1996; Pannitto, Kahane, et al., 2026)

1/ What is Universal Dependencies?

Universal Dependencies (UD) is a cross-linguistic framework for annotating grammatical structure in corpora (Marneffe et al., 2021).

Its basic idea is simple:

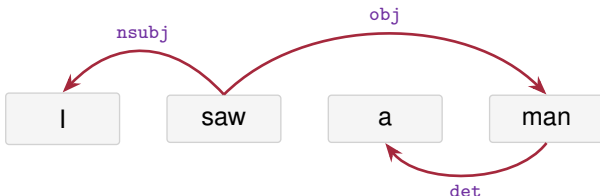
- ✓ each sentence is represented as a **dependency tree**
- ✓ words are connected through relations such as `nsubj`, `obj`, `obl`, `advmod`
- ✓ each word also receives a part-of-speech tag and morphological features
- ✓ the same inventory of relations is used across languages as much as possible

I saw a man → `nsubj(saw, I)` `obj(saw, man)`

» UD makes grammatical structure searchable and comparable across corpora and languages.

1/ What is a dependency?

A dependency is a **directed relation** between two words.



Head = organizing word

Dependent = contributing word

Label = grammatical function

The dependency relation says: **this word contributes syntactically to that word.**

This lets us represent syntax as a network of word-to-word relations, rather than as nested phrases.

1/ The problem for UD

Universal Dependencies was developed primarily on written data, where syntactic dependencies are normally represented within sentence boundaries (Dobrovoljc, 2022; Marneffe et al., 2021).

Spoken data make this assumption unstable:

- ✓ speakers stop, restart, overlap and repair
- ✓ one syntactic structure may be distributed across turns
- ✓ feedback may occur inside another speaker's ongoing unit
- ✓ segmentation may follow prosody, turn organization or syntax

»»» The question is not whether spoken data fit UD perfectly, but how to encode speech-specific phenomena while remaining UD-compatible.

1/ Why still use UD?

UD was not designed for spoken interaction. But this makes it a useful shared framework for testing what spoken data require and what they highlight.

Shared format	spoken data can be compared with written data and across languages
Computability	interactional phenomena become searchable, countable and testable
Conversion	speaker-based and dependency-based views can coexist
Pressure test	spoken language reveals the limits of UD and points to necessary changes and extensions

»»➔ The aim is not to force speech into UD, but to see how far a shared dependency framework can be extended without losing interactional structure.

1/ Maximal unit segmentation

Spoken treebanks differ in how they segment units.

- Prosodic approach** segment at prosodic completion; often close to interactional boundaries, but difficult to apply consistently
- Syntactic approach** link material when a syntactic dependency can be identified; risks producing very long units
- Turn-based approach** preserve speaker production and turn organization; may hide dependencies across speakers

Guiding principle: if you can link, link.

» Segmentation is already an analysis of what counts as syntactically connected in speech.

(Kahane et al., 2021; Pietrandrea et al., 2014)



1/ Two views of the same interaction

For spoken interaction, one view is often not enough.

Speaker-based view	keeps each speaker's utterance as a separate tree; preserves production, turn organization and metadata
Dependency-based view	reconstructs the syntactic unit across speakers; makes cross-turn dependencies searchable as regular dependencies

speaker turns → Coconstruct/Backchannel pointers → dependency tree

» The aim is not to erase interactional organization, but to make conversion between views possible.

(Pannitto, Kahane, et al., 2026)

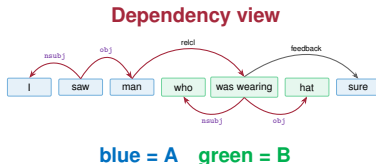


1/ Two views: a simple picture

The same exchange can be represented in two complementary ways.



preserves speakers and turns



» Speaker view asks: *who said what, in which turn?* Dependency view asks: *which words belong together syntactically?*

2/ What is a co-construction?

We consider as **co-constructions** syntactic relations that hold between utterances produced:

- ✓ by different speakers
- ✓ by the same speaker after an interruption by another participant

The central diagnostic is syntactic:

Can the second utterance be linked to an exact token in the previous syntactic structure?

If yes, the second utterance is not syntactically autonomous in the interactional sequence.

(Pannitto, Kahane, et al., 2026)



2/ A first typology

When a co-construction involves two speakers, two broad configurations are relevant.

A. Same dependency slot

Speaker B realizes a syntactic dependency that is already projected or already realized in Speaker A's structure

B. New dependency

Speaker B adds a syntactic dependency that was not already part of Speaker A's tree

A.1 Completion

an unrealized dependency is completed

A.2 Parallel realization

the same slot is realized by both speakers, often in overlap

A.3 Subsequent realization

a completed slot is repeated, corrected or reformulated

B Extension

a new modifier or dependent is added

»» The distinction concerns the status of the syntactic slot at the moment of the second contribution.

2/ A.1 Completion

»» Speaker A leaves a syntactic dependency unrealized, and Speaker B provides the missing material.

L his position is **pretty uh**
A **stable.**

A she has a **very nice**
B **attitude**

- ✓ the dependency is projected but incomplete
- ✓ the responding token realizes the missing syntactic head or dependent
- ✓ the speaker-based tree needs a way to mark incompleteness

2/ Completion in Italian

BO146 ((ride)) vabbè te ti ci ti ci devo **fare pe::r**

well I have to make you by

BO139 ((ride)) **mestiere**

trade

Scrap=Yes

marks the unfinished material in the speaker-based view

Promotion=case

records that the promoted preposition would normally be
case

Coconstruct=obl::...

links the second contribution back to the projecting token

»➔ The annotation preserves both facts: the first unit is unfinished, and the second unit completes a dependency relation.

2/ A.2 Parallel realization

»»➔ The same syntactic dependency is realized by both speakers, often with overlap.

BO118 c'è architettura::, c'è::: [ingegneri~]

BO140 [ingegn~ ingegn]eria: sì

BO118 sì che è [molto buona] anche

there is architecture, there
is engineer-
enginee- engineering yes
yes, which is very good too

- ✓ Speaker A projects a slot
- ✓ Speaker B starts filling the same slot
- ✓ Speaker A may also complete it

»»➔ Overlap is not only a timing problem. It may reveal shared access to the emerging syntactic trajectory.



2/ A.3 Subsequent realization

»» Speaker A completes a syntactic structure; Speaker B then targets the same slot again.

PKP126 quindi l- la linea **tra finzione e realtà**

cioè **tra verità non verità**

non ho ancora capito [dove sta]

so the line between fiction
and reality

that is between truth and
non-truth

I still have not understood
where it lies

PKP125 [più tra de]tto e non detto

more between said and
unsaid

conj:reform

captures reformulation: different denotations for the same
discourse target

Coconstruct=Yes

keeps the cross-speaker status visible after conversion



2/ B. Extension

»»➔ Speaker B does not fill the same slot. Speaker B adds a new dependency to Speaker A's structure.

BO139	quando si parlano sopra	when they speak one over the other
	devi mettere delle parentesi quadre	you have to put square brackets
	intorno alle parole	around the words
BO146	ah okay	oh okay
BO139	che si sovrappongono	that overlap
BO136	come nei sottotitoli	like in subtitles

»»➔ The second contribution expands the tree: it contributes new syntactic material licensed by the earlier structure.

2/ Repair and reformulation

Co-constructed syntax also interacts with false starts and repairs.

BO147 infatti te potresti fare **il sotto~**

quello che fa i sottotitoli

BO145 **sottotitolatore**

indeed you could be the
subti-
the one who does subtitles
subtitler

reparandum

regular UD relation for abandoned/repared material

repair

speaker-based helper relation when the later item must
carry the co-construction pointer

conj:reform

for reformulations, confirmations and corrections that stack
on the same denotation



3/ Beyond co-construction

Co-constructions show that syntactic dependencies may cross turns and speakers.

But spoken interaction also contains material that does not complete a syntactic slot, while still shaping the unfolding structure.

Co-construction	adds, completes or reformulates syntactic material
Backchannelling	supports another speaker's ongoing contribution
Overlap	adds a timing layer to the syntactic sequence
Question-answering	fills projected slots, but with a different sequential status

»»➔ The next block moves from shared syntax to the wider interactional architecture around it.

3/ Backchannelling

Backchannels are short productions uttered while another participant holds the floor.

To be recognized as backchannels, they must:

- ✓ be addressed to the content of another speaker's contribution
- ✓ not be required or expected by the preceding turn
- ✓ not require a reaction from the main speaker

`discourse:backchannel` in dependency view
`Backchannel=<sent-id>::<tok-id>` in speaker view

(Mereu, Cangemi and Grice, 2024; Ward and Tsukahara, 2000)



3/ Backchannel: example

BO145 ma perché mamma c'ha dei
pregiudizi nei miei confronti

BO139 **mhmh**

BO145 e poi daniela non devi avere
pregiudizi su di me

but because mum has
prejudices against me

mhmh

and then Daniela you
should not have
prejudices about me

- ✓ the main speaker keeps the floor
- ✓ the backchannel validates reception/alignment
- ✓ the speaker-based unit remains separate
- ✓ the dependency view inserts it into the temporal/syntactic flow



3/ Backchannel vs. answer

Not every short response is a backchannel.

Backchannel	not projected as an answer; does not claim the floor
Polar reply	expected by the question; answers the previous turn
wh-answer	fills an interrogative slot; may be analyzed as co-constructed syntax in some frameworks

BO145 [(non) prendi l'insalata?]

will you not take the salad?

BO139 [sì sì]

yes yes

» The difference is sequential: expected answer vs. supportive response during another trajectory.

3/ Wh-questions

Answers to *wh*-questions are close to co-construction because they instantiate a slot opened by the interrogative structure.

TOR001 **dove** vai ad arrampicare?

where do you climb?

TOI007 **al bi side** vicino alla colletta

at the Bside close to
Colletta

30 Magalie utilise depuis combien de temps notre
boule magique ?

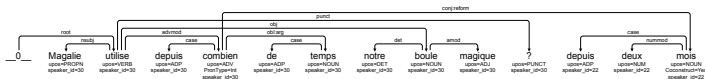
How long has Magalie
been using our magic ball?

22 depuis deux mois

for two months

»➡ The open question is whether to treat this as co-construction proper or as a distinct question-answer configuration.

3/ Question-answer in dependency view



➡ The answer *depuis deux mois* is linked to the interrogative element and marked as Coconstruct=Yes.

3/ Overlap

Overlap introduces a second dimension into the representation of spoken syntax.

Speaker-dependent who produced each unit, and in which speaker trajectory

Time-dependent which units were simultaneous, delayed, interrupted or inserted

```
Overlap=<sent-id>[:<tok-start>:<tok-end>]
```

- ✓ token ids can mark overlapping spans
- ✓ multiple overlap values may be needed for more than two speakers
- ✓ timing and syntax should remain independently searchable



4/ What should annotation make visible?

The risk is to treat co-construction as an annotation problem only.

But the deeper issue is representational:

- ✓ syntax is visible in the words and dependency relations
- ✓ interactional work is distributed across speakers, timing and uptake
- ✓ corpora must preserve both dimensions if they are to model spoken grammar

The goal is not only to annotate co-constructions, but to represent the interactional architecture that makes them possible.

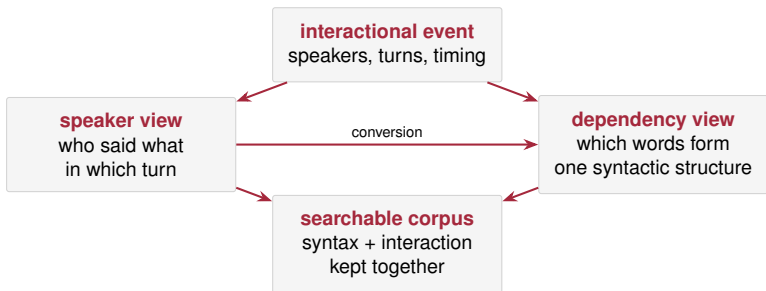
4/ From infrastructure to representation

The annotation must therefore keep track of three layers at once.

Visible syntactic structure	dependency relations, repaired or expanded material
Interactional architecture	speaker identity, turn organization, overlap, feedback, projected completion
Conversion path	the possibility of moving between a speaker-based and a dependency-based representation

»»➔ A spoken treebank is useful only if the syntactic tree does not erase the interactional supports underneath it.

4/ A minimal architecture



- ✓ speaker view preserves production and participation
- ✓ dependency view preserves syntactic connectedness
- ✓ conversion preserves the relation between the two

4/ The useful technical core

Only a small technical core is needed to make the architecture explicit.

Coconstruct pointer	links a later unit to the earlier syntactic governor
Backchannel pointer	links feedback to the unit it validates or supports
speaker_id at token level	keeps speaker identity after dependency-based conversion
Overlap, Scrap, Promotion	preserve timing, incompleteness and projected syntactic role

These features are not just labels. They are anchors that keep visible syntax connected to its interactional infrastructure.

4/ Implementation as proof of concept

The proposal is already being tested in two resources:

SUD French Rhapsodie speaker-based and dependency-based views

UD Italian speaker-based and dependency-based views,
KIParlaForest based on spoken KIParla data

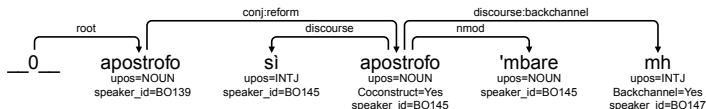
KIParlaForest testbed	Size
Tokens	9,135
Sentences	1,007
Sentences with backchannels	134
Co-construction annotations	70

»»➡ The point of implementation is not only technical validation: it tests whether the interactional architecture can remain visible in computable data.

(Pannitto, Kahane, et al., 2026; Pannitto, Zucchini, et al., 2025)



4/ What the tree preserves



Syntactic continuity

Interactional scaffolding

the dependency tree keeps together elements that belong to the same emerging construction
speaker information and feedback show how that construction was supported, interrupted or resumed in interaction

»»» The tree is not only a final syntactic product: it can also preserve traces of the interactional architecture that made it possible.

4/ Beyond co-construction

Once the architecture becomes visible, co-construction is only one case in a broader representational problem.

Backchannels

feedback that supports another speaker's ongoing trajectory

Overlap

simultaneity that may or may not interrupt syntactic projection

Pauses and non-verbal behavior

temporal and embodied supports for segmentation and interpretation

Metadata

speaker, recording, alignment and speech-event information needed for comparison

»»➔ The general question is how to represent spoken interaction without flattening it into written-like sentences.

Take-home points

1. Spoken syntax is **incremental**, temporal and recipient-designed.
2. Co-constructions show that syntactic dependencies may cross speaker and turn boundaries.
3. Speaker-based segmentation preserves interactional production; dependency-based representation preserves syntactic connectedness.
4. UD can encode much of this if it preserves the interactional architecture beneath the dependency tree.
5. The broader challenge is collective: spoken treebanks need shared practices.

Opening the discussion

Across the three lectures, we have moved from interaction as infrastructure to meaning-making and, finally, to syntactic representation.

The general question is now broader than any single phenomenon:

What kind of grammar becomes possible if we take interaction, reciprocity and co-construction as part of the structure to be described?

- ▶ What should be represented in the annotation?
- ▶ What can be left to interpretation, metadata or corpus context?
- ▶ Where do current frameworks help, and where do they need to be extended?

» Questions, doubts and counterexamples are part of the empirical work.

Thank you

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